Characterization of Message Lengths for Capacity-Achieving Linear PIR

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***Abstract*—We characterize the possible message lengths of capacity-achieving linear private information retrieval (CALPIR) schemes. Let p be a prime power. Let a finite field $\mathbb{F}_p$ be the message alphabet. Let $M, N \geq 2$ be integers. By taking direct sums of previous CALPIR constructions with message lengths $L = N-1$ and $L = N^{M-1}$, we obtain a CALPIR scheme for every message length $L = a(N-1)+bN^{M-1}$, where a and b are nonnegative integers with $(a,b) \neq (0,0)$. We prove the converse: no other message lengths are possible for CALPIR schemes.**

Fix p, L, M, N. The base of all logarithms is p. Each server $n\,(\in [N] := \{1,\dots,N\})$ stores the same M messages $W^1,\dots,W^M\,(\in \mathbb{F}_p^L)$. The user's objective is to retrieve W^θ while keeping all information about the value of θ perfectly private from each server. Server n receives a query $Q_n\,(\in \mathcal{Q}_n)$ from the user, where $\mathcal{Q}_n$ is a finite set. It sends an answer $A_n\,(\in \mathbb{F}_p^{L_{\mathrm{A}}})$ to the user. The user attempts to recover W^θ from θ, $Q_1,\dots,Q_N$, and $A_1,\dots,A_N$. Here, $W^1,\dots,W^M$, $Q_1,\dots,Q_N$, $A_1,\dots,A_N$, and θ are random variables satisfying the following conditions.

M messages $W^1,\dots,W^M$ are i.i.d. and uniformly distributed over $\mathbb{F}_p^L$. The desired message index θ is uniformly distributed over $[M]$. $\mathrm{I}\left(\theta, Q_{[N]}\ ;\ W^{[M]}\right) = 0$, where $Q_{[N]} := (Q_1,\dots,Q_N)$, $W^{[M]} := \left((W^1)^\top,\dots,(W^M)^\top\right)^\top$ (Query-Independence). For every $n \in [N]$, $\mathrm{H}\left(A_n|Q_n, W^{[M]}\right) = 0$ (Functionality). For every m, $\mathrm{H}\left(W^m|A_{[N]}, Q_{[N]}, \theta = m\right) = 0$ (Correctness). For every n, m, m', q, $\Pr(Q_n = q|\theta = m) = \Pr(Q_n = q|\theta = m')$ (Privacy).

A PIR scheme is called capacity-achieving if its download cost $\sum_{n\in[N]} \mathrm{H}\left(A_n|Q_n\right)$ attains the lower bound $L\frac{1-(1/N)^M}{1-(1/N)}$ [1]. A PIR scheme is called linear if, for every $n \in [N]$, there exists a collection of deterministic matrices $\{\mathbf{G}_n^q : q \in \mathcal{Q}_n\}$ such that $A_n = \mathbf{G}_n^{Q_n} W^{[M]}$.

Theorem 1. *Let p be a prime power and let $M, N \geq 2$ be integers. A CALPIR scheme with message length $L \geq 1$ exists if and only if there exist nonnegative integers a, b such that $L = a(N-1)+bN^{M-1}$, $(a,b) \neq (0,0)$.*

Tian et al. [1] proved that the minimum CALPIR message length is $N-1$, whereas Theorem 1 characterizes all possible message lengths.

Proof. Achievability follows by combining, via direct sums, the CALPIR construction for $L = N^{M-1}$ [2] and that for $L = N-1$ [1]. We prove only the converse. Fix a CALPIR scheme with message length L, an index $m \in [M]$, and a query realization $\mathbf{q} \in \left\{\, \mathbf{q}' \in \prod_n \mathcal{Q}_n \;\middle|\; \Pr(Q_{[N]} = \mathbf{q}'|\theta = m) > 0 \,\right\}$. By Properties P1 and P2 in [1, Theorem 2][1], for every n,

$$\mathrm{H}\big(A_{[N]}|Q_{[N]} = \mathbf{q}, W^m, \theta = m\big) = \mathrm{H}\big(A_n|Q_{[N]} = \mathbf{q}, W^m, \theta = m\big)\,,$$

$$\begin{aligned}&\mathrm{H}\left(A_{[N]}|Q_{[N]} = \mathbf{q}, \theta = m\right)\\ &= \sum_n \mathrm{H}\left(A_n|Q_{[N]} = \mathbf{q}, \theta = m\right).\end{aligned}$$

Therefore,

$$\begin{aligned}&N\mathrm{H}\left(A_{[N]}\middle|Q_{[N]} = \mathbf{q}, W^m, \theta = m\right)\\ &= \sum_{n\in[N]} \mathrm{H}\left(A_n\middle|Q_{[N]} = \mathbf{q}, W^m, \theta = m\right)\\ &\leq \sum_{n\in[N]} \mathrm{H}\left(A_n\middle|Q_{[N]} = \mathbf{q}, \theta = m\right)\\ &= \mathrm{H}\left(A_{[N]}\middle|Q_{[N]} = \mathbf{q}, \theta = m\right)\\ &= L + \mathrm{H}\left(A_{[N]}\middle|Q_{[N]} = \mathbf{q}, W^m, \theta = m\right). \qquad (1)\end{aligned}$$

Moreover,

$$\begin{aligned}&\mathrm{H}\left(A_{[N]}|Q_{[N]} = \mathbf{q}, W^m, \theta = m\right)\\ &= \mathrm{H}\left(W^m, A_{[N]}|Q_{[N]} = \mathbf{q}, \theta = m\right) - L.\end{aligned}$$

By linearity and the uniform distribution of $W^{[M]}$, $\mathrm{H}\left(W^m, A_{[N]}|Q_{[N]} = \mathbf{q}, \theta = m\right)$ is equal to the rank of the linear map from $W^{[M]}$ to $(W^m, A_{[N]})$. Therefore, $\mathrm{H}\left(A_{[N]}|Q_{[N]} = \mathbf{q}, W^m, \theta = m\right)$ is a nonnegative integer. Since $A_n = \mathbf{G}_n^{Q_n} W^{[M]}$ and $W^{[M]}$ is independent of $(\theta, Q_{[N]})$, averaging Eq. (1) and using Privacy and Correctness yields $\sum_n \mathrm{H}(A_n|Q_n) - L = \mathrm{H}(A_{[N]}|Q_{[N]}, W^m, \theta = m) \leq u$, where $u := \left\lfloor \frac{L}{N-1} \right\rfloor$. Thus, $L\frac{1-(1/N)^M}{1-(1/N)} - L \leq u$. We set $b := L - u(N-1)$ and $a := u - b(1 + N + \cdots + N^{M-2})$. Then $0 \leq b < N-1$, and substituting $L = u(N-1)+b$ into the last inequality gives $b(N^{M-1}-1) \leq u(N-1)$. Thus, a is a nonnegative integer, and $L = a(N-1)+bN^{M-1}$. □

References

[1] C. Tian, H. Sun, and J. Chen, "Capacity-achieving private information retrieval codes with optimal message size and upload cost," *IEEE Trans. Inf. Theory*, vol. 65, no. 11, pp. 7613–7627, Nov. 2019.

[2] H. Sun and S. A. Jafar, "Optimal download cost of private information retrieval for arbitrary message length," *IEEE Trans. Inf. Forensics Security*, vol. 12, no. 12, pp. 2920–2932, Dec. 2017.

[1] Linear PIR is uniformly decomposable. Removing dependent answer rows querywise preserves correctness and entropy and identifies expected answer length with conditional-entropy download cost. Thus [1, Theorem 2] applies.